



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	SOLID STATE PHYSICS				
Type of Course	Core in Major				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	The prerequisites for a course in solid state physics typically include a strong foundation in classical mechanics, electromagnetism, quantum mechanics, thermodynamics and statistical mechanics, and optionally solid state chemistry, along with recommended physics laboratory experience				
Course Summary	In a course on solid state physics, students delve into the fundamental principles governing the behavior of matter in its solid phase, exploring topics such as crystal structures, electronic properties, thermal properties, magnetic phenomena, and their applications, with an emphasis on understanding the microscopic origins of macroscopic properties and phenomena observed in solid materials.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the principles of crystal structures and their classification schemes	Understanding	Conceptual Knowledge	Written exams, quizzes
CO2	Analyze the electronic band structure of solids and its implications for electrical conductivity	Analyzing	Procedural Knowledge	Problem sets, simulations
CO3	Explain the principles of quantum mechanics as applied to solid state systems	Understanding	Conceptual Knowledge	Class discussions, presentations
CO4	Predict and interpret the thermal properties of solids using statistical mechanics	Applying	Conceptual Knowledge	Laboratory experiments, projects
CO5	Investigate the magnetic properties of materials based on their atomic and electronic structures	Analyzing	Procedural Knowledge	Research papers, presentations
CO6	Apply solid state physics principles to real-world applications such as semiconductor devices	Applying	Procedural Knowledge	Case studies, group projects
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	CRYSTAL STRUCTURE AND BINDING		13	20
	1	Periodic array of atoms, fundamental types of lattices	2	
	2	Index systems for crystal planes, simple crystal structures	2	
	3	Diffraction of waves by crystals, scattered wave amplitude, Brillouin zones	4	

	4	Crystal binding: Crystals of inert gas, ionic crystals, covalent crystals, metals,	3	
	5	Hydrogen bonds	2	
	Pages 3 – 18, 27 – 40, 51 – 72 of Book 1			
II	FREE ELECTRON THEORY AND BAND THEORY		13	20
	6	Free electron Fermi gas: Energy levels in 1D, Effect of temperature on FD distribution	3	
	7	Free electron gas in 3D, heat capacity of electron gas	2	
	8	Electrical conductivity and Ohm's law, motion in magnetic field	2	
	9	Thermal conductivity of metals	2	
	10	Energy bands: nearly free electron model	2	
	11	Bloch equations, Kronig Penney model	2	
	Pages 135-159, 165-171 of Book 1			
III	SEMICONDUCTOR AND SUPERCONDUCTIVITY		12	20
	12	Band gap, equations of motion	2	
	13	Intrinsic carrier concentration	2	
	14	Impurity conductivity	2	
	15	Introduction to superconductivity, Sources of superconductivity, Response of magnetic field, Meissner effect	2	
	16	Origin of band gap, Isotope effect	1	
	17	Elements of BCS theory	1	
	18	Normal tunnelling and Josephson effect, High T _c superconductivity	2	

	Pages 189 – 216, Chapter 8 of Book 1 Sections 17.1 – 17.4, 17.6, 17.7, 17.11, 17.13, 17.14 of Book 2			
IV	CRYSTAL VIBRATIONS AND THERMAL PROPERTIES		7	10
	19	Vibrations of crystals with monatomic basis, First Brillouin zone	2	
	20	Group velocity, Long wavelength limit	1	
	21	Two atoms per primitive basis, quantisation of elastic waves	2	
	22	Phonon heat capacity (qualitative idea only) mention Debye and Einstein model	2	
	Respective Sections of chapter 4 and 5 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Y and σ -Interference Method (a) elliptical (b) hyperbolic fringes. To determine Y and σ of the material of the given specimen by observing the elliptical and hyperbolic fringes formed in an interference setup.		
	2	Y & σ by Koenig's method.		
	3	Dielectric constant by Lecher wire - To determine the wavelength of the waves from the given RF oscillator and the dielectric constant of the given oil by measurement of a suitable capacitance by Lecher wire setup.		
	4	Constants of a thermocouple and temperature of inversion.		
	5	Susceptibility measurement by Quincke's and Guoy's methods - Paramagnetic susceptibility of salt and specimen.		
	6	Conductivity, Reflectivity, sheet resistance and refractive index of thin films.		
	7	Hall effect in semiconductors - To determine the carrier concentration in the given specimen of semiconducting material.		

	8	Absorption spectrum of KMnO_4 and Iodine - To determine the wavelength of the absorption bands of KMnO_4 and to determine the dissociation energy of iodine molecules from its absorption spectrum.		
	9	Ionic conductivity of KCl / NaCl crystals.		
	10	To study the Thermoluminescence of F-centres of Alkali Halides.		
	11	Variation of dielectric constant with temperature of a ferroelectric material (BariumTitanate) .		
	12	Dielectric constant of a non polar liquid.		
	13	Ultrasonic interferometer - To determine the velocity and compressibility of sound in liquids.		
	14	Band gap energy of Ge by four probe method - To study bulk resistance and to determine band gap energy.		
	15	Determination of Band gap energy of Ge and Si using diodes.		
	16	Thermionic work function - To determine the thermionic work function of the material of the cathode of the given vacuum diode/triode from the characteristic at different filament currents.		
	17	Simulate the temperature dependent heat capacity of different metals/solids with known Debye temperatures, such as Pb ($\Theta_D = 88 \text{ K}$), Gd ($\Theta_D = 169 \text{ K}$), Ag ($\Theta_D = 215 \text{ K}$) and KCl ($\Theta_D = 308 \text{ K}$), in thermal equilibrium at various temperatures using the Debye Model. Use a normalized axis for comparison and draw the conclusions (see example 24.2 in Book 3).		
	18	Simulate the density of states (degeneracy) of a one dimensional, two dimensional and three dimensional non interacting system (See section 6.2 in Book 4)		
	19	Simulate the equation of state (isotherms) for a van der Waals gas (see section 26.1 in Book 3).		
	20	Simulate the temperature dependent heat capacity of free electron gas for various electronic number density (See page 141, chapter 6 of Book 1).		
	21	Simulate the phonon dispersion relation in a solid		
	22	Simulate the response function, amplitude and average energy as function of driving frequency for a damped harmonic oscillator for various damping (see example 33.5 in Book 3).		

	23	Simulate the solution of the Kronig-Penney model for periodic potential in solid.		
	24	Simulate the electrical conductivity and hole concentration as a function of electron concentration for a semiconductor at a fixed temperature for $n_p = \text{constant}$ (See page 214, chapter 8 of Book 1).		

Books and References:

1. Introduction to Solid State Physics by Charles Kittel; Wiley India Edition (Book 1)
2. Solid State Physics: Structure and properties of materials by M.A.Wahab (Third Edition)
3. Concepts in Thermal Physics (Oxford Edn., 2006) by Stephen J Blundell and Katherine M. Blundell.
4. Thermal Physics Tutorials with Python Simulations (CRC Press, 2023) by Minjoon Kouh and Taejoon Kouh.
5. Solid State Physics by Neil W. Ashcroft and N. David Mermin.
6. Solid State Physics: Essential Concepts by David W. Snoke.
7. The Oxford Solid State Basics by Steven H. Simon.
8. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html>

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	0	0	3	1	3	1	1	0	3	1	0
CO 2	3	2	2	0	3	1	3	1	1	1	3	1	0
CO 3	3	0	1	0	3	1	3	1	1	0	3	1	0
CO 4	3	2	2	1	3	1	3	1	1	1	3	1	0
CO 5	3	2	2	0	3	2	3	1	1	1	3	1	0
CO 6	3	1	3	2	3	2	3	2	2	1	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	